WELCOME TO TERM THREE OF 2024 AQUACULTURE COURSE (SA220)





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FEED FORMULATION

- There are several methods o f formulating feeds and these are Pearson's Square M ethod, Algebraic Equation , Trial and Error, and Linear Programming .
- The Pearson's Square method is recommended in formulating a supplemental feed with only two to four ingredients.
- The Trial and Error method is generally used in calculating a formula for a complete diet with many ingredients.

FEED FORMULATION

- In commercial feed production where cost is a principal consideration , a computer program (Linear Programming) is used to combine feed ingredients that will give an effective formulation with the lowest cost.
- Mathematical calculations using these methods, except for Linear Programming, are given in the following examples.

Pearson’s Square and Algebraic Equation Methods

- **E xample 1.** When only two feed ingredients are to be combined.
- To determine the amount of each ingredient in a supplemental feed that would contain 25% protein using Fish Meal (FM) and Rice Bran (RB), with protein contents of 63% and 10%, respectively.
- **Pearson’s Square Method:**
- a) First draw a square and write the desired protein level of the feed at the center of the square.

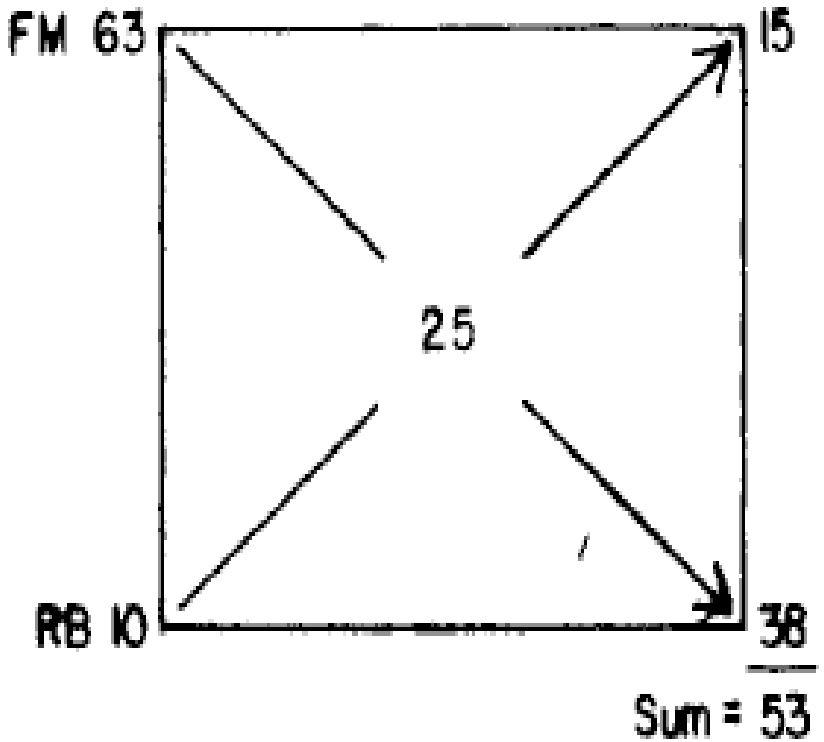
Pearson’s Square and Algebraic Equation Methods

- b) Write the two ingredients with their respective protein contents on each comer of the left side of the square, the ingredient with higher protein on the upper and with lower protein on the lower left hand corner.
- c) Subtract the desired level of protein from each ingredient and write the difference on the right corner of the square that is diagonally opposite the protein level of each ingredient.

Pearson’s Square and Algebraic Equation Methods

- d) Get the sum of the numbers at the right side of the square.
- e) Determine the percentage of each ingredient needed for the feed formula by dividing the numbers written on the right hand side by the sum of the difference multiplied by 100.

Pearson's Square and Algebraic Equation Methods



$\frac{15}{53} \times 100 = 28.3\%$ FM

$\frac{38}{53} \times 100 = 71.7\%$ RB

Pearson’s Square and Algebraic Equation Methods

- About 283 g FM and 717 g of RB are required to make a kg of the supplemental feed that will contain 25% protein.
- To check for the amount of protein in the feed, the protein content (in percent) of each ingredient is multiplied by the amount of FM and RB (283 g and 717 g respectively) to be combined.
- The amount of protein coming from each ingredient is added to get the total amount. The total amount is divided by 1000 and multiplied by 100 to give 25% protein.

Pearson's Square and Algebraic Equation Methods

FM	=	283 g x 63%	=	178.3 g protein
RB	=	717 g x 10%	=	71.7 g protein
		Total protein		250.0 g /kg or 25% protein

or (250 / 1000) x 100 = 25%

Pearson's Square and Algebraic Equation Methods

- **Algebraic Equation Method:**

Let: x = g FM/ kg feed
y = g RB/ kg feed

so that :

$x + y = 1000$ g feed (Equation I)

$0.63x + 0.10y = 250$ g protein/1000 g feed (Equation II)

Multiply Equation I by 0.10:

$0.10x + 0.10y = 100$ g (Equation III)

Subtract Equation III from Equation II:

$$\begin{array}{rl} 0.63x + 0.10y & = 250 \text{ g protein/1000 g feed} \\ -(0.10x + 0.10y & = 100 \text{ g}) \\ \hline 0.53x + 0 & = 150 \end{array}$$

$x = 150/0.53 = 283$ g FM

Substitute in Equation I:

$283 + y = 1000$

$y = 1000 - 283 = 717$ g RB

Pearson’s Square and Algebraic Equation Methods

- Example 2: When three or more feed ingredients are available for use.
- Formulate a fish diet to contain 35% protein by combining the following ingredients with their respective protein content:
- FM - 60% protein
- Soybean Meal (SBM) - 45% protein
- RB 8% protein
- Com Meal (CM) 12% protein

Pearson’s Square and Algebraic Equation Methods

- Assume the proportion of 1 part FM to 2 parts SBM or 1:2 and equal parts of RB and CM or 1:1.
- FM and SBM are the main sources of protein while RB and CM are the main sources of carbohydrate or energy but also contain protein.

Pearson’s Square and Algebraic Equation Methods

- Pearson’s Square Method:
- a) Draw a square and write at the center the desired protein level as in the first example.
- b) Calculate the protein level from the protein sources FM and SBM. According to the specified ratio: 1:2
- FM : 1 x 60 = 60
- SBM : 2 x 45 = 90
- Average : 150 / 3 = 50

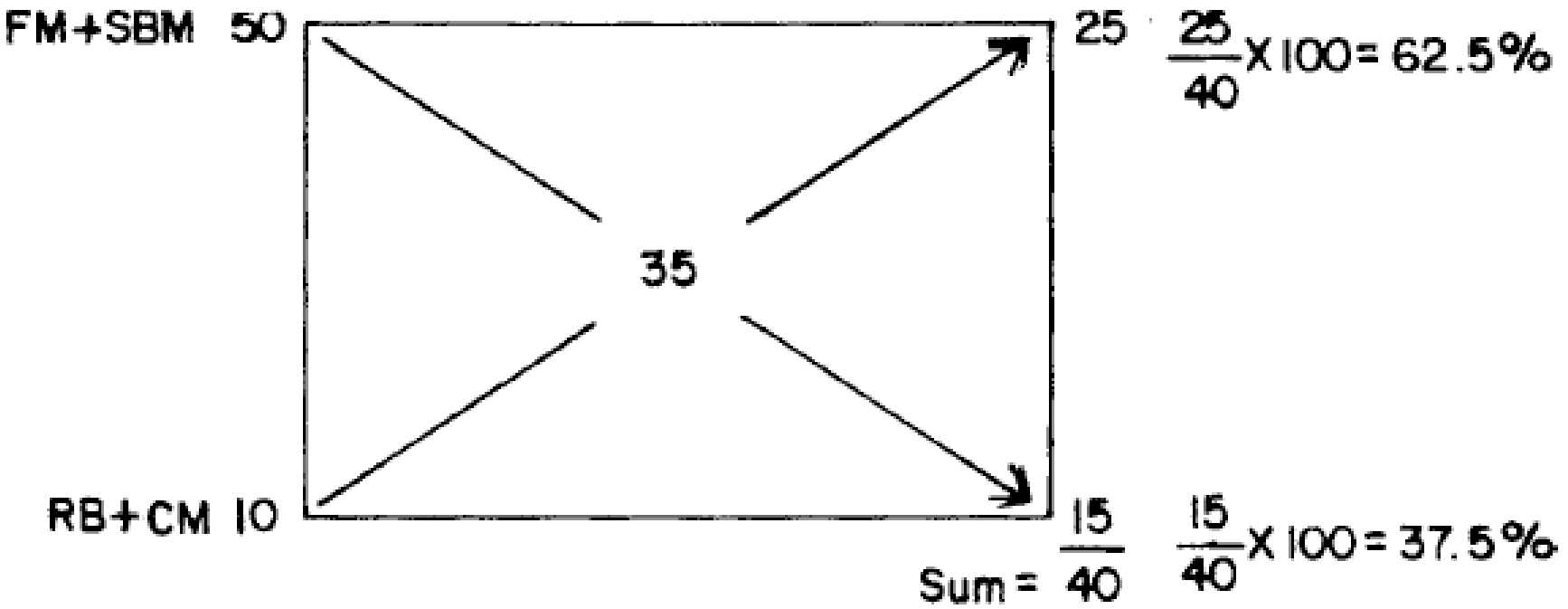
Pearson’s Square and Algebraic Equation Methods

- c) Calculate the energy sources, RB and CM also according to the ratio 1:1
- RB : 1 x 8 = 8
- CM : 1 x 12 = 12
- Average : 20 / 2 = 10
- d) Write the calculated average protein content on the upper left hand corner for the protein sources and the protein content of the energy sources on the lower left corner of the square.

Pearson’s Square and Algebraic Equation Methods

- e) Write the desired protein level at the center and subtract this value from the protein content of FM and SBM and protein content of RB and CM.
- Write the number diagonally opposite the ingredients or on the lower right hand corner for protein sources and upper right hand corner for energy sources.

Pearson's Square and Algebraic Equation Methods



Pearson’s Square and Algebraic Equation Methods

- f) Multiply the final percentage derived for the protein sources by 1/3 for FM and 2/3 for SBM.

For the energy sources, multiply by 1/2 each of the RB and CM to find out the exact amount of each ingredient to be used in the formula with the desired level of 35% protein

Pearson's Square and Algebraic Equation Methods

Protein sources	=	62.5%				
				In %		In g/kg feed
FM	=	62.5 x 1/3	=	20.83%		208.3
SBM	=	62.5 x 2/3	=	41.67%		416.7
Energy sources	=	37.5%				
RB	=	37.5% x 1/2	=	18.75%		187.5
CM	=	37.5% x 1/2	=	18.75%		187.5
Total				100.00%		1,000

g) To check that a kilo of feed contains 35% protein, proceed as follows:

FM	=	208.3 g x	60% protein	=	125.0 g protein
SBM	=	416.7 g x	45% protein	=	187.5 g protein
RB	=	187.5 g x	8% protein	=	15.0 g protein
CM	=	187.5 g x	12% protein	=	22.5 g protein
Total per 1000 g feed					350.0 g protein/kg or 35%

END

THANK YOU